Exploring Tannin-Furanic Rigid Foam Insulation Research and Marketability

Bio-Based Insulation Alternatives and Optimization Using Tannin Extract

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An Abstract of the Tannin Extract Applications With A Focus On Rigid Foam Research in Comparison To Market Leaders

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Abstract

This paper focuses on the newer “green” tannin-furanic rigid foam insulation, comparing it with commercially available alternatives and other bio-based materials. It will cover the extraction process, focusing on traditional liquid/solid extraction and supercritical fluid extraction, and provide an overview of current applications in construction and other markets, such as construction adhesives and the fashion industry. The case study results explain the rigid foam's characteristics:, use, application, ease of production, and testing properties, such as thermal conductivity, porosity, compression, and mechanical strength, showing that tannin-furanic foams perform comparably to market-leading competitors, such as polyurethane. The material is lightweight, easy and quick to prepare, and can be fully adjusted to be up to 100% bio-based materials. These findings underscore the material's promise for advancing sustainable building practices and commercial viability.

Research Question:

HOW ARE TANNIN-FURANIC RIGID FOAMS BEING OPTIMIZED AND PROMOTED AS A COMMERCIALLY VIABLE, BIO-BASED ALTERNATIVE TO PETROLEUM-BASED INSULATION MATERIALS LEADING THE MARKET?

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1. Introduction

1.1 – Background

Construction companies have set forth the image of chemical foam insulation, which has taken over the market. Materials such as polyurethane, polystyrene, or phenolformaldehyde have exceptional properties such as thermal and sound insulation while being lightweight; however, their inherent defects then get overwritten–for example, their high flammability, brittleness, acidity, along with their petroleum-based properties. While modifiers do exist, it is likely at the cost of another carbon-heavy product–such as polylols, which have raised further concerns due to their high toxicity and harmful compounds released in the air upon burning. Such issues have been brought to light in recent years amid this evolving climate in the construction industry. As traditional work technologies shift toward greener options, the insulating properties have become a highly relevant issue.

In pursuit of this sustainable design, insulation has become a top priority for climate protection and heat transfer, reducing the need for AC, heating, and other materials with a higher carbon footprint. Most insulation materials are chosen for their ability to reduce heat flow by conduction, keeping energy inside the ventilated home. Though this is often at the hands of advertised multifaceted industry options, leaving petroleum compounds within the walls of our homes. Bio-based alternatives have been tested in market procedures at favorable and equal levels from algae, moss, and tannin-furanic foam types.

Tannin-furanic rigid foams, in particular, have attracted increasing interest thanks to their easy preparation and lightweight construction. The foam product formulation has a 90% minimum biomass content, with self-polymerization enabled by a furfuryl alcohol co-reaction, giving it the “spray foam” expansion type we typically see in most homes. It exhibits properties comparable to those of market counterparts, including lightweight, thermal insulation, and improved flame-retardant performance. There are ongoing research projects to further increase this biomass content and to limit the 10% non-biomass product, thereby reducing the small amounts of Volatile Organic Compounds (VOCs) produced during growth. This means chemical compounds are released into the air, harming the environment and possibly human health. Though most other commercial products share this same downside, it becomes an investment issue, especially when researching a greener alternative.

1.2 – Testing Parameters For Insulation Materials

Thermal conductivity is the rate at which heat passes through a material. In insulating materials, it measures the resistance to heat transfer, serving as an indicator of structural behavior and energy efficiency. This value, typically denoted by "k," indicates performance: a lower value signifies better insulation. The thermal conductivity can vary with density and material structure. For example, dense, porous, and fibrous materials usually have lower thermal conductivity than solid, softer materials.

Thermal conductivity is typically measured using a standardized setup. The material sample is placed between two metal plates at different temperatures—one heated, one cooled. A sensor, such as a transiently heated plane sensor or a thermocouple, is attached to the sample to measure temperature changes and heat flux as the system reaches equilibrium. The steady-state value yields the material's thermal conductivity, or "k." The current market leaders, such as polystyrene and polyurethane, hold conductivity values ranging from approximately 0.03 to 0.01.

In insulation materials, uniform foam cell distribution ensures even porosity while maintaining structural strength. Small, well-distributed cell windows provide a strong skeletal framework, enhance compressive strength, and promote consistent thermal conductivity. This uniformity can be achieved with modifiers or by ensuring homogeneous mixtures during foaming. On the molecular scale, structural uniformity depends on cross-linking during curing, which is often analyzed using X-ray photoelectron spectroscopy. This technique reveals transitions in ion valence states and the formation of new states during oxidation.

Apparent density, pulverization ratio, and compressive strength are important metrics for comparing the thermal conductivity of insulation materials and determining their appropriate applications. These measurements correspond to the foam formulation and reflect macroscopic cross-linking. Materials with similar densities demonstrate electrostatic binding and reinforced cell structures, which are crucial for insulation performance and longevity. Lower values in these metrics indicate better insulation performance, higher durability, and improved resistance to aging.

1.3 – Research Objectives

Tannin-furanic rigid foams are being optimized as commercially viable, bio-based alternatives to petroleum-based insulation materials through innovative formulation and

processing techniques. Tannins, which are natural, widely available biocompounds, provide a renewable and sustainable raw material source. Recent advancements focus on enhancing the mechanical properties, thermal insulation performance, and fire resistance of tannin-furanic foams to match or exceed those of petroleum-based products. Industrial research emphasizes the elimination of harmful blowing agents, improving uniformity of cell structure, and increasing compatibility with existing manufacturing processes. These efforts promote tannin-furanic foams as high-performance, environmentally responsible insulation solutions suitable for large-scale commercial applications, thus positioning them as leaders in the transition toward sustainable building materials.

This project specifically investigates optimizing tannin-furanic rigid foams for use in insulation by developing formulations that do not require traditional blowing agents. Adaptations are being explored to ensure ease of application and integration into current construction practices. The aim is to deliver a green, high-performance insulation material that maintains mechanical integrity and thermal efficiency, thereby supporting the commercial adoption of Tannin-furanic foams as a sustainable alternative to conventional petroleum-based insulation.

2. Extraction and Contents of Tannin

Tannins are polyphenolic compounds found naturally in vegetal cells, though their familial connections range across all sorts of vegetation. The main polyphenolic metabolites are found in bark, stems, seeds, roots, buds, or leaves, but can also be found in fruits such as grapes, blackberries, or walnuts. They are a defensive agent protecting their host plant from fungi, pathogens, insects, and animals. They also possess the unique ability to connect with multiple functional groups, enabling stable cross-linking with molecules such as proteins or carbohydrates.

Typically, tannins are divided into two categories of polyphenols: hydrolysable and condensed. Hydrolysable tannins are composed of ellagic and gallic acids with a sugar core and thus have far more limited sources in nature. Condensed tannins are flavonoids lacking a sugar core and are the dominant type in the world market, accounting for over 90% of commercial availability. They are most commonly used in the leather industry for hide treatment and their red/brown color. It has also been adapted as an adhesive in the wood and insulation industry. Further extracts have explored medicinal properties and

clothing markets-these markets are fully explored below under “Existing Application of Tannin Extracts”.

The most traditional method of tannin extraction is through solid/liquid extraction. This involves a solvent passing through and pulverizing excess tissue, including chlorophylls and lipids. The second step is the extraction of tannin from the particulate matter. Since its solubility varies depending on the plant from which it is extracted, different solvents may be used, such as water, ethanol, acetone, or methanol. Tannins have limited solubility with polar solvents, thus ethanol and water are the most common choices–water being the simplest and most efficient on an industrial scale as it can extract 29-887 mg/g of dried extract depending on the original plant source. It is the most economical and environmentally friendly choice, though it requires a large amount of water, high temperatures, and long time periods (24-48 hours) to properly extract the tannins. However, the use of polar solvents has significantly increased thermal stability and expedited extraction, reducing the time to nearly 4.5 hours.

Supercritical extraction is a secondary method in which substances are brought to a supercritical state and subjected to the highest temperature and pressure at their critical point to achieve gas-liquid equilibrium. The solvent for this is most commonly carbon dioxide, as it is readily available, non-toxic, and nonpolar. Other polar co-solvents may be added to optimize flow rate, extraction time, moisture content, and the required pressure and temperature (as the most important parameters).

Other extraction methods include pressurized water extraction, microwave-assisted extraction, and ultrasound-assisted extraction.

3. Existing Applications of Tannin Extracts

Tannin extract, as its research emphasis suggests, has applications based on environmental implications. One of the most prominent examples of this is in its work as a coagulant. This is used in the treatment of surface and industrial wastewater, where it destabilizes colloids and suspended solids. It can break down inorganic compounds derived from aluminum salts, such as ferric chloride or aluminium sulfate. However, vegetal-based coagulants serve as an alternative to metal-based ones, as they consume less alkalinity and are renewable. Some studies have already marketized and applied the treatment, with comparable, and sometimes even better, performance than conventional options. For example, the tannin from Acacia mearnsii (a flowering plant native to

Australia) has been used as a commercially viable coagulant, sold as “Tanfloc SG,” through chemical modification and synthetic additives, which have optimized its maximum output and antibiotic properties. In urban effluents, it showed improved performance compared to synthetic options and was capable of reducing the load of prokaryotic organisms. It has also been applied to the dairy industry with no significant difference in colour, turbidity, chemical oxygen demand (COD), or total solids (TS) removal.

Another application of tannin extract is as an adhesive in wood, tire, and concrete products. Like the tannin rigid foam reaction, many of these applications rely on the polymerization of formaldehyde. Strong tannin adhesives require about a 10% equivalent in the mix, which introduces a level of toxicity to the bio-based adhesive, as mentioned above. However, increasing research is being done on replacing furfuryl alcohol and glyoxal in water. Though it increased the procedure time, the results show similar reactions and applications for wood and insulation adhesive materials, while improving hydrophobicity by 12%. In concrete, it is also used as a plasticizer, reducing the water content in the mix. It lengthens the setting time and early development strength, but contributes to the concrete's fluidity without compromising its setting.

Other applications include flotation agents for industrial processing, in the fabric industry (leather, dyes, food additives, etc.), and in medicine through antioxidant, antimicrobial, anthelmintic, antiviral, and anti-inflammatory applications.

4. Experimental Procedures

4.1 – Case Study Framework

The following case studies exemplify the growing interest and efforts to market tannin-furanic rigid foams in the construction industry. They experimentally demonstrate the material's strength and ease of production, and they modify and optimize it for different uses. The first case study uses the base control mix to demonstrate product repeatability, ease of mixing, and performance as a bio-based alternative. The following sources build upon the control group using modifiers to find rational mixture designs for a given experimental response. The prediction systems based on relative proportions and formulations develop environment-friendly tannin-furanic foams with simple characterization of defined features. To demonstrate the effectiveness and commercial

potential of these innovations, the following case studies provide detailed examples of tannin-furanic rigid foam applications in various real-world stages.

4.2 – A Case Study From "Natural Tannin-based Rigid Foams as Insulation for Doors and Wall Panels" by G. Tondi, A. Pizzi, and R. Olives

Under the case study "Natural Tannin-Based Rigid Foams as Insulation for Doors and Wall Panels" by G. Tondi, A. Pizzi, and R. Olives, this paper examines the potential applications of natural tannin-based rigid foams through experimental testing and product repeatability through a set of identical samples tested under different conditions of weathering and aging. Thanks to tannin's natural fire resistance and its preexisting use in insulation adhesives, its catalytic reduction supported research into its sole insulating strength.

The researchers prepared 10 foam samples within plywood boxes (50 × 50 × 70 mm). The foaming process occurred in three stages: mixing, expansion, and curing. During the mixing phase, all components—tannin extract, formaldehyde, furfuryl alcohol, additives (as needed), and water—were blended until homogeneous. Expansion began after the addition of a strong acid, initiating an exothermic self-condensation reaction among the furfuryl alcohol, tannin, and formaldehyde. This reaction caused the blowing agent to evaporate and foam the mixture, which stabilized within minutes as curing occurred. The final product, 2 hours post-curing, is shown in Figure 1.

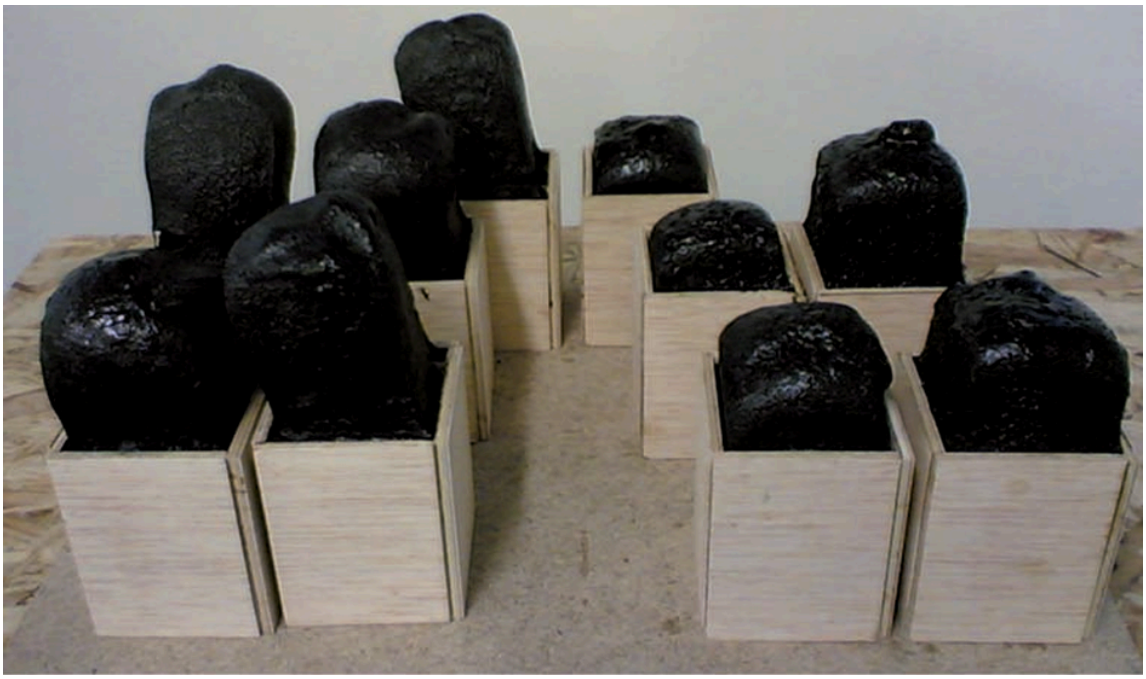


Figure 1 *– Appearance of cured and hardened test samples for exterior weather exposure prior to cutting*

The exact composition included 5.2 g of furfuryl alcohol, 3.7 g of 37% formaldehyde solution, 3 g of water, and 15 g of Acacia mearnsii tannin extract. The acid catalyst comprised 1.5 g of diethyl ether and 6.0 g of toluene-4-sulphonic acid, added and mixed for 10 seconds before foaming began. The foam temperature rose to about 40°C, producing a final density of 0.05–0.1 g/cm³.

After drying, the samples were cut uniformly and subjected to natural weathering for 1 day, 4 months, and 8 months to assess performance under environmental exposure. Thermal conductivity was measured using two metal plates held at 60°C and 10°C, respectively, with samples and thermocouples placed between them. After 2 hours, steady-state readings showed that the tannin-based foams—comprising roughly 95% natural materials—exhibited thermal conductivities comparable to those of synthetic polymer foams such as polyurethane. These values are shown in Table 1 below. As we can see, from the 1-day sample (DEE3) to the 8-month weathered sample (DEE4), there is little variation or correlation among the values, though all lie within standard thermal conductivity requirements for market competition.

Sample	Direction	Density (g/cm^3)	Thermal Conductivity (W/m.K)
DEE2	Growth	0,115	**0,26**
DEE3	Growth	0,091	**0,30**
DEE4	Growth	0,056	**0,27**
DEE2	Perpend.	0,115	**0,24**
DEE3	Perpend.	0,0914	**0,26**
DEE4	Perpend.	0,056	**0,23**

Table 1 *– thermal conductivity values for tannin-based foams of different density in the direction and perpendicular to the direction of foam growth*

Two key findings emerged: (1) the tannin foams provided effective insulation, and (2) they demonstrated strong resistance to humidity and weathering. Even after 8 months of exposure, the foams did not bond to or damage the acidic wood surfaces—a common issue with polymer foams. The structure remained stable and undamaged, highlighting its suitability for use in exterior and humid conditions.

In conclusion, this study showed that tannin-based rigid foams present a sustainable, bio-derived alternative to oil-based insulators, with comparable performance and promising applications in hollow-core wooden frames and weather-exposed areas.

4.3 – A Case Study From the "Optimisation of "green" tannin-furanic foams for thermal insulation by experimental design" by C. Delgado-Sánchez a, F. Santiago-Medina , V. Fierro a, A. Pizzi b c, A. Celzard

This paper conducts 14 tests to identify the best thermal insulators with the highest mechanical resistance using green tannin-furanic foam. Using the tannin extract, furfuryl alcohol, and catalyst as variables, across all samples, the foams exhibited extremely high porosity—between 95 and 98%—with bulk densities ranging from 0.022 to 0.069 g/cm^3. These values demonstrate that the foaming process produces lightweight cellular structures whose final cure is highly sensitive to ingredient ratios, without the need for formaldehyde, which is a VOC and a harmful compound.

The first generation of tannin-furanic rigid foams used the exothermic reaction involving 5% formaldehyde, a carcinogenic, highly flammable VOC. This study aimed to replace this with “greener” materials, increasing the bio-based content from 90-95% to 98%. It did so by replacing the formaldehyde and blowing agent with a solution of furfuryl alcohol and ethylene glycol. The experimental design then focused on three formulation parameters: tannin, furfuryl alcohol, and a catalyst. All other variables were held constant to isolate their effects on thermal conductivity, compressive strength, and elastic modulus—key response variables—and to facilitate normalization within the prediction system. The effect of furfuryl alcohol on the reaction has not yet been normalized. During foaming, furfuryl alcohol increases heat generation and disrupts the curing process, causing rapid expansion and a temperature gradient from the core to the exterior. This phenomenon negatively impacts the structure and uniformity of the final sample.

The final bulk densities were measured at a constant compression rate of 0.0025 s-1, and the thermal tests were analyzed along the foaming direction. The results are shown in Table 2.

No.	ρ_b (g cm^{-3})	ρ_s (g cm^{-3})	ρ_r	θ (%)	Compressive strength (MPa)	Elastic modulus (MPa)	Thermal Conductivity (W m^{-1} K^{-1})
1	0.025	1.420	0.018	98.2	0.035	0.90	0.038
2	0.024	1.416	0.017	98.3	0.019	0.34	0.038
3	0.069	1.417	0.049	95.1	0.175	5.55	0.045
4	0.041	1.443	0.028	97.2	0.042	0.53	0.044
5	0.030	1.432	0.021	97.9	0.030	1.00	0.037
6	0.060	1.417	0.042	95.8	0.118	1.50	0.041
7	0.032	1.423	0.022	97.8	0.023	0.42	0.040
8	0.040	1.446	0.027	97.3	0.048	0.80	0.042
9	0.024	1.449	0.016	98.4	0.023	0.60	0.038
10	0.034	1.445	0.023	97.7	0.022	0.33	0.038
11	0.031	1.450	0.022	97.8	0.026	0.86	0.041
12	0.030	1.415	0.021	97.9	0.031	0.56	0.039
13	0.026	1.435	0.018	98.2	0.015	0.28	0.041
14	0.022	1.433	0.015	98.5	0.020	0.50	0.039

Table 2 – *Experimental results for the 14 foam formulations*

The thermal performance of the materials was similarly dependent on composition to the compression strength, as predicted. Measured thermal conductivities ranged from

0.037 to 0.045 W/m·K, placing the foams within the range expected for commercial insulating materials. Lower-density foams with higher porosity produced lower thermal conductivity values, while denser foams exhibited slightly higher conductivity. This trend reinforces the trade-off in foam design: improving insulation generally requires increasing porosity, but doing so can sacrifice mechanical robustness.

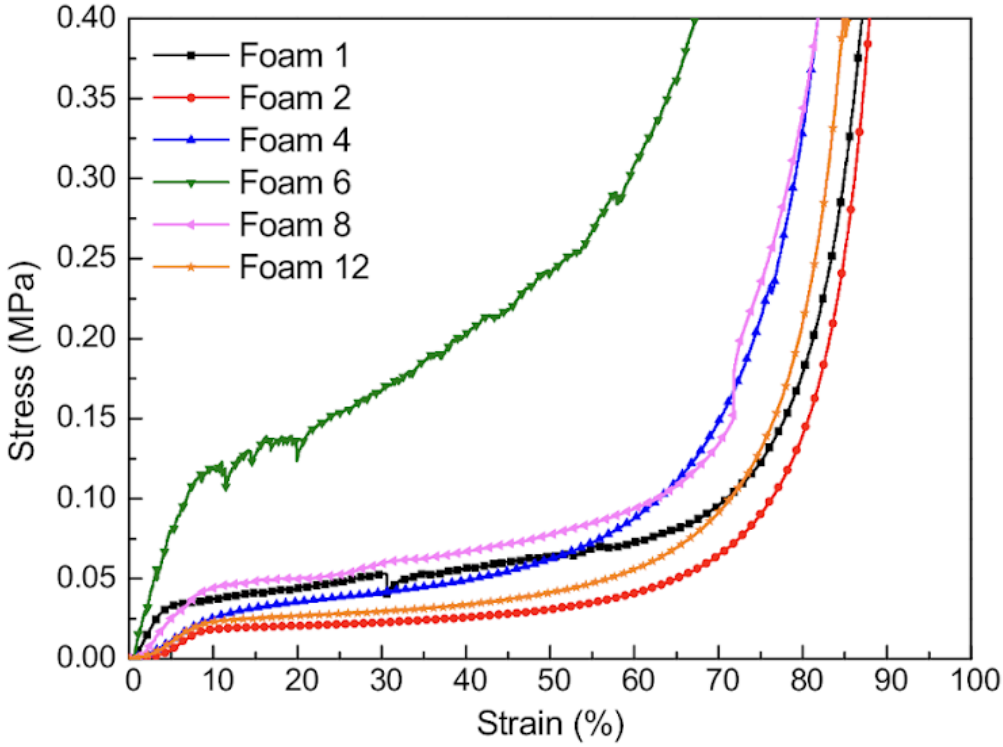


Figure 2 *– Stress-strain curves of tannin-furanic foam samples*

Mechanical testing revealed that the foams primarily behaved as elastic–plastic materials, a contrast to the brittle failure often seen in tannin-based foams cured with formaldehyde. Compressive strengths spanned a wide range—from 0.015 to 0.175 MPa—and elastic moduli ranged from 0.28 to 5.55 MPa, with both properties increasing proportionally to density. Denser foams showed a more variable stress–strain profile (Figure 2), indicating the onset of brittle fracture within individual cell layers, whereas low-density foams deformed much more smoothly. The study also showed strong consistency and reproducibility: two formulations prepared identically (#9 and #14) produced nearly identical thermal and mechanical properties.

Overall, this case study shows that tannin-furanic foams exhibit insulating performance comparable to that of synthetic polymer foams and mechanical properties

sufficient for many construction applications. Their combination of low thermal conductivity, tunable density, and high bio-based content makes them strong candidates for sustainable insulation applications. In this case study, we have demonstrated our ability to secure a top thermal insulation candidate, moving from 95% green content, similarly seen in section 4.2, to 100% bio-based material by removing formaldehyde. It also provided a tool for optimizing the antagonistic properties of porosity and mechanical strength through a prediction model suitable for the three main ingredients: tannin, furfuryl alcohol, and a given catalyst. The model tests demonstrated properties consistent with the prediction model while developing environment-friendly tannin-furanic foams with the desired features for industrialization.

4.4 – A Case Study From the "Metal Ions Fortified Tannin-Furanic Rigid Foam: The Impact on the Uniformity and Mechanical Performance" by Yang Yang, Haishu Wu, Jun Shang, Faijan Li, Bertrand Charrier, Hisham Essawy, Antonio Pizzi, Xiaojian Shou, and Xinyi Chen

The study focuses on the modifiers of the furfuryl alcohol tannin-furanic foam mixture, similar to the case study in section 4.3. The limited sustainability of non-sustainable blowing agents and their limited mechanical properties due to brittleness and application constraints encouraged research into potential modifiers. This includes successful applications, the avoidance of foaming agents, and the addition and replacement of nanocellulose and wood cellulosic fibres to maintain the same foaming pore size and improve compression strength.

The team uses a commercial mimosa-condensed tannin extract from Brazilian acacia trees, furfuryl alcohol, formaldehyde, FeCl3, CuCl2, and C4H6O4Zn purchased from Shanghai Macklin Biomedical Corporation. Using the Mixing expansion and curing method explained in section 4.2, they used a 2-hour post-foaming process followed by an oven heating at 80 degrees Celsius overnight for final curing and solidification.

Bulk density was measured as a ratio of weight to geometric volume of equally cut samples. Compression strength was calculated as the maximum force before 10% deformation divided by the area of the foam sample; this was repeated five times per mix. The pulverisation ratio, for the determination of friability, was measured by pulling a piece of sandpaper along at a constant speed of 10 mm/s and weighing it with a 200 g

weight for 30 back-and-forth tests. Finally, thermal conductivity was measured and repeated to assess thermal stability using X-ray photoelectron spectroscopy.

In Figure 3, we show microCT images of the tannin-based foams without a blowing agent. We see that the control group (Figure 4a) is more porous and uneven than the copper-ion-enhanced samples. The fine, uniform structure in the others demonstrates stable foam cell structures that enable more stable, reproducible air entrapment, helping avoid fusing. Furthermore, the skeletal structure shows far fewer irregular defects, especially over the large cell wall “windows”. This defect can compromise the integrity of the structure and, by extension, its overall strength.

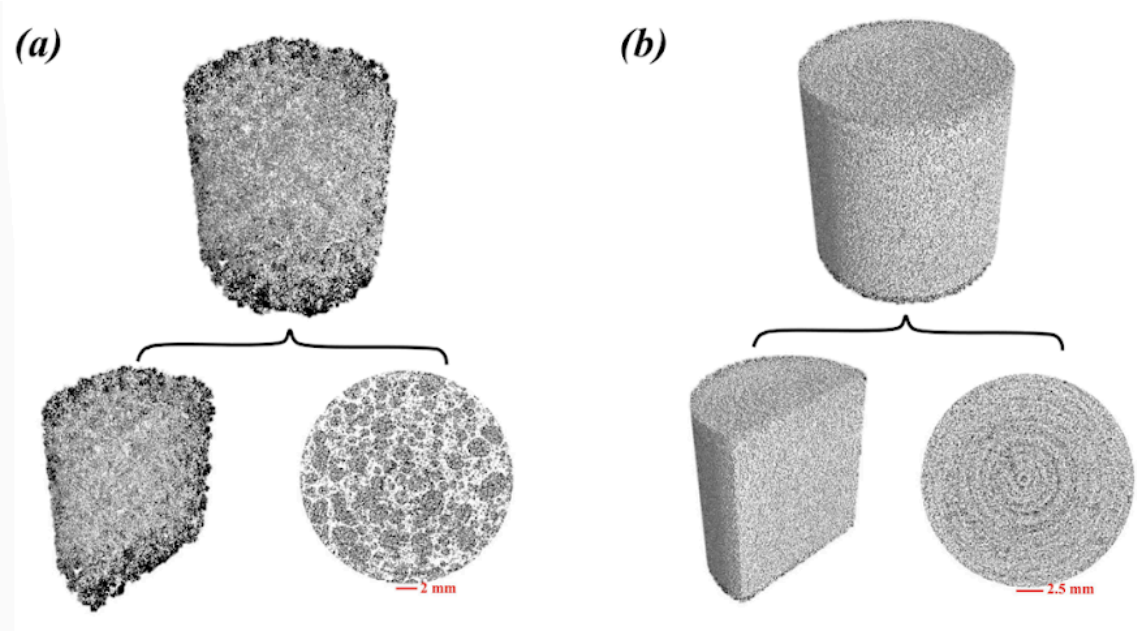


Figure 3 – *Micro-CT images of control sample (a) and the copper metal ion-enhanced (b) sample internal structures of the longitudinal cross section*

In compression tests, the pulverisation ratio further confirmed the positive role of the inserted metal ions. The control sample showed a 28.86% pulverisation rate, whereas the modified foams showed 7.57-11.01% (a reduction of 61-73%). This means there were more intact foam cells per unit volume, and the cell pillars were far less likely to collapse under greater stress. The thermal stability, decomposition, and conductivity showed similar stages and trends to those of the control group (seen in Figure 4).

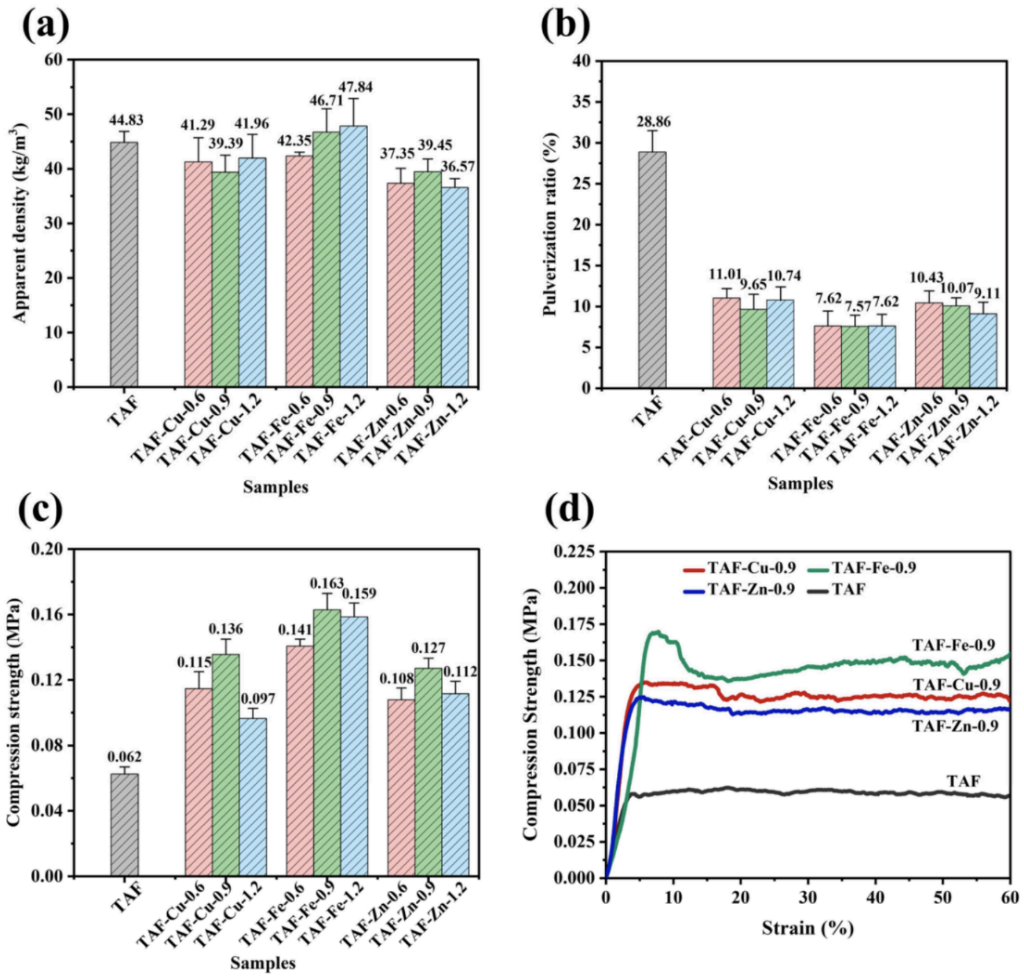


Figure 4 – *The apparent density (a), pulverization ratio (b), compression strength (c), and stress-strain curves (d) of the control group TAF and metal ion-enhanced tannin-furanic foams*

In summary, the upgrading approach with metal ions copper, iron, and zinc displayed a more uniform performance and structure, which promoted compression strength, pulverisation ratio, and a robust network. This implies significant application potential, as it not only competes with traditional bioinsulation materials on mechanical grounds but also with other commercial materials. It has a lower cost (in moderation) and offers opportunities for broader applications in construction, anti-collision materials, transportation, and exterior wall insulation.

5. Discussion and Conclusions

The growing interest in developing sustainable, innovative materials has led to a resurgence of bio-based building materials. Tannin-furanic rigid foams leverage the existing extract market and a widely available resource to become an active competitor to the current insulation leaders – polyurethane foams. Rather than using petroleum-based materials, inorganic crops, bio-insulation manufacturing offers equivalent versatility, using metal ion modifiers and other bio-material modifiers we saw in the case studies, a low density, good mechanical resistance, and simple processing. It is not only flame-retardant but also reduces smoke emissions and completely neutralizes the toxic gases emitted by current insulator leaders, such as carbon monoxide and hydrogen cyanide, providing high potential in the civil construction field. As the industry continues to prioritize sustainability, tannin-furanic rigid foams are well-positioned to lead the transition toward greener building solutions.

The series of case studies listed above demonstrated the materials' strength, ease of production, optimization methods with bio-based modifiers, increased prediction model accuracy, and effectiveness compared with petroleum-based market leaders. Through case study one, the weathering's longstanding thermal conductivity showed market-level conduction, with no acidity to the wood it was held in, unlike petroleum-acid-based alternatives, which slowly tear away at our timber. For the second, we can see an increase in biobased content without loss of strength, and accurate predictive modeling for mixture optimization. This consistency enables industrial applications tailored to system needs. Finally, the third optimized the system, similar to polylol adhesion modifiers, for other spray foams by polar ion enhancement during the curing process. The increased uniformity also enabled heavier construction product applications, such as industrial use.

Incorporating biocomposite insulations such as tannin-furanic foams represents a significant advancement toward sustainable construction and green building initiatives. Continued optimization and global adoption of these materials will expand access to eco-friendly solutions and promote responsible building practices worldwide.

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